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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION - MARCH 2026

FOURTH SEMESTER

DISCIPLINE SPECIFIC ELECTIVE (DSE) - MG4DSECFS200 - VIRTUAL PRIVATE NETWORK SECURITY

(2024 ADMISSION ONWARDS)

Part A (2 Marks Each)

1. Simplify the reasons for using VPN in companies.

- **Secure Remote Access:** Allows employees to connect safely to the company's internal private network from any location via the public internet.
- **Data Confidentiality & Cost Savings:** Encrypts sensitive corporate data streams securely, replacing the need for highly expensive physical leased lines.

2. Define VPN protocols.

VPN protocols are a standardized framework of cryptographic rules, encapsulation techniques, and transmission agreements (such as IPsec, SSL/TLS, PPTP, and L2TP) that dictate exactly how data is packaged, encrypted, and securely tunneled over public paths.

3. Define 2FA.

Two-Factor Authentication (2FA) is an identity management security verification mechanism that requires users to provide two separate, distinct components of evidence before acquiring VPN access (typically something they know, like a corporate password, and something they possess, like an automated smartphone OTP token).

4. Explain why AES-256 is preferred in VPN.

- **High Security Tier:** Employs a 256-bit cryptographic key depth, creating an exponentially massive combination field that renders mathematical brute-force breaches completely impossible.
- **Optimized Performance:** Provides supreme hardware execution efficiency, delivering robust encryption defenses without significantly reducing active network bandwidth speeds.

5. Explain about VPN server attacks.

These represent malicious, targeted efforts aimed at crashing or hijacking a central network VPN Gateway. Common variants include **Denial of Service (DoS/DDoS)** flooding attacks designed to crash infrastructure reachability, and direct software vulnerability exploitations to extract administrative privileges.

6. State any two features of TLS.

- **Cryptographic Confidentiality:** Secures channel environments using robust, negotiated symmetric session keys.
- **Data Transmission Integrity:** Utilizes a Message Authentication Code (MAC) architecture to confirm payload packages are not intercepted or structurally modified during transmission.

7. Define RBAC.

Role-Based Access Control (RBAC) is a systems security approach where network resource privileges are explicitly mapped to predefined job profiles within the company (e.g., HR, engineering, accounts) rather than being configured manually on an individual user basis.

8. Describe the role of authentication in VPN.

Authentication serves as the initial gatekeeper function that formally verifies the actual credentials and explicit identity of any remote entity or system applying to enter the tunnel network, mitigating unauthorized vector entry.

9. List the main steps in VPN connection setup.

1. **Initiation & Parameter Handshake:** Client contacts the endpoint gateway; the nodes negotiate matching cryptographic specs.
2. **Identity Evaluation & Tunneling:** User parameters/certificates are validated, symmetric crypto elements are safely configured, and a secure operational virtual pipeline is locked down.

10. Identify the steps involved in 3DES encryption process.

The 3DES protocol applies standard Data Encryption Standard (DES) mechanics sequentially three times to every specific text block utilizing separate keys via an **Encrypt → Decrypt → Encrypt (EDE)** functional process flow.

11. Define the key steps in user authentication.

1. **Credential Collection:** The system collects identity tokens (passwords, PINs, certificates, or biometrics) submitted by the remote endpoint client.
2. **Directory Validation & Allocation:** The gateway validates submission packets against active directories (like RADIUS/LDAP) and returns user authorization permissions.

12. Explain Phase-1 negotiation in IPsec.

Phase-1 establishes a secure administrative management path. The two communication endpoints authenticate each other and negotiate a foundational secure framework called an **ISAKMP/IKE Security Association (SA)** using a Diffie-Hellman cryptographic exchange.

Part B (Short Essay Type - 5 Marks Each)

13. Differentiate between Man-in-the-Middle attack and eavesdropping attack.

Feature	Eavesdropping Attack	Man-in-the-Middle (MitM) Attack
Attack Nature	Passive security threat framework.	Active, invasive security threat framework.
Core Mechanism	Silently monitors and records channel packets without structural disruption.	Intercepts traffic, with the ability to alter, inject malicious code, or drop packets.
Primary Objective	Stealthy extraction of confidential secrets, credentials, or corporate intel.	Session hijacking, identity impersonation, or injection of fraudulent traffic.
Detection Difficulty	Extremely high, as normal operational flows are left entirely unhindered.	Relatively easier via latency changes, structural anomalies, or certificate errors.

14. Explain the operations performed in IKEv2 and write the applications of IKEv2.

Operations Performed:

- Streamlines tunnel configuration by utilizing a reduced 4-message structural baseline exchange compared to IKEv1.
- Coordinates the foundational `IKE_SA_INIT` message phase to lock down crypto preferences and Diffie-Hellman attributes.
- Executes the `IKE_AUTH` routine to verify endpoint identities and safely construct the initial IPsec Child Security Association.
- Integrates native **MOBIKE** (Mobility and Multihoming Protocol) protocols to cleanly navigate dynamic user address handovers.

Applications: Primary configuration profile engine for enterprise remote worker access paths, highly dynamic smartphone network VPN clients, and clean infrastructure Site-to-Site pipelines.

15. Describe NAC in VPN environments.

Network Access Control (NAC) performs strict health posture evaluations on remote connection terminals before admitting their traffic into the corporate VPN network ecosystem. The system verifies key parameters, including active OS upgrade baselines, enabled endpoint antimalware engines, and operational device firewalls. Devices that fall short of these metrics are quarantined into isolation zones, effectively protecting local internal data networks against malware contamination.

16. Write the steps to deploy remote access VPN.

1. **Gateway Setup & Initial Configuration:** Define public interface interfaces, configure specific subnet routing rules, and pool internal IP resources for remote connections.
2. **Authentication Core Integration:** Bind the central edge firewall framework to corporate directory servers (LDAP/RADIUS) and apply multi-factor access challenges.
3. **Security Rule & Policy Mapping:** Define precise ACL parameters, select traffic routing styles (e.g., split-tunneling vs. full-tunneling), and establish session lifecycles.
4. **Endpoint Package Provisioning:** Output and install preconfigured endpoint connection tools or security profiles across remote terminal groups.
5. **Validation Testing & Logging Activation:** Run verification testing with sample user subsets and align event reporting pipelines to a central SIEM server.

17. Compare Remote Access VPNs and SSL/TLS VPNs.

- **Remote Access VPNs (IPsec-driven):** Operate at the Network layer (OSI Layer 3). They typically demand dedicated endpoint software suites and assign full virtual physical interfaces to the client terminal, granting them wider network infrastructure access.
- **SSL/TLS VPNs:** Function at the Application layer (OSI Layer 7). They offer a flexible, browser-centric connection model requiring no dedicated software installation, limiting user interactions specifically to designated web applications rather than the entire internal network.

18. Discuss about encryption algorithms used in VPNs.

- **DES (Data Encryption Standard):** Legacy 64-bit symmetric structural cipher running an old 56-bit working key length. Deprecated due to vulnerability to brute-force computational decoding.
- **3DES (Triple DES):** Wraps data blocks with the standard DES structure sequentially three times. Offers stronger security than DES but suffers from high latency and low computational processing efficiency.
- **AES (Advanced Encryption Standard):** The modern global standard symmetric block cipher. Highly efficient, secure, and performant across low-power devices. Extensively scaled inside modern VPN networks at 128, 192, and 256-bit key layers.

19. Explain the components of RADIUS architecture.

- **RADIUS Client / Network Access Server (NAS):** The edge access control device (such as an internal corporate VPN gateway) that collects user requests and bridges them over to the central AAA database.
- **RADIUS Server:** The centralized processing hub that evaluates incoming connection validation packets, checks credentials, and returns authorization codes.
- **User Profile Directory Database:** The backend enterprise repository (e.g., Active Directory or LDAP environments) containing specific access records, authorization rules, and group mappings.

20. Explain logging and monitoring techniques for VPN management.

- **Centralized Syslog Infrastructure:** Routinely gathers operational connection events, authentication tracking, connection durations, and failure warnings to a central SIEM matrix.
- **SNMP Monitoring Tracks:** Collects real-time performance metrics, including system resource usage (CPU/Memory), overall bandwidth traffic levels, and concurrent active tunnels.
- **Security Audit Appraisals:** Systematically reviews authorization access trails to detect malicious credential-stuffing patterns or anomalous connection spikes.

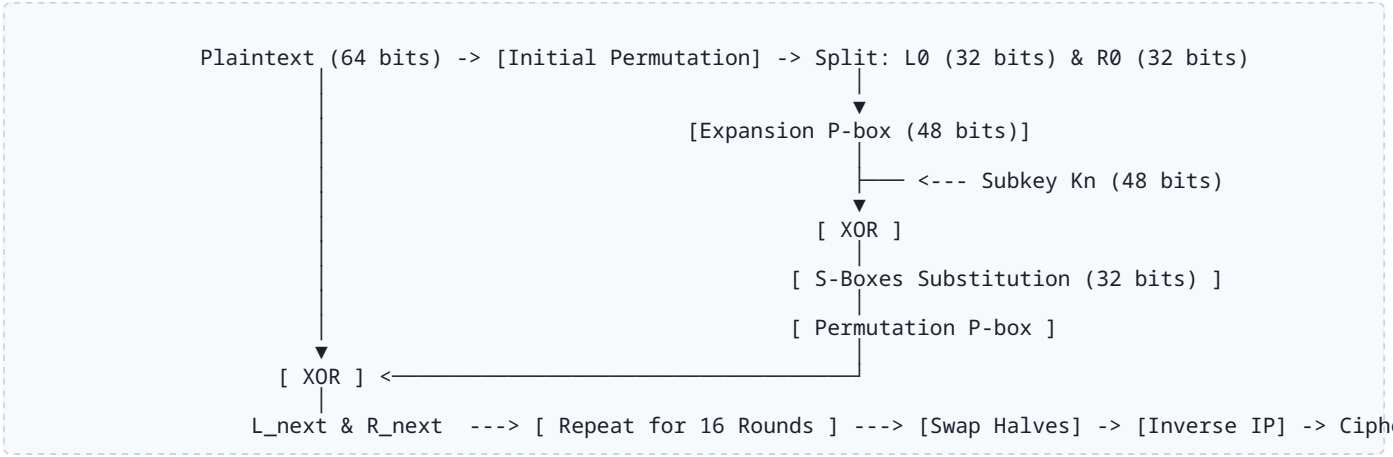
Part C (Essay Type - 10 Marks Each)

21. Explain in detail the DES encryption process with all 16 rounds and diagram.

Core Architectural Attributes: The Data Encryption Standard (DES) is a symmetric block cipher based on the Feistel network architecture. It processes data payloads in fixed blocks of 64 bits using a 64-bit original key length (of which 56 bits serve as the active cryptographic key, while the remaining 8 bits function as parity verification points).

Step-by-Step Processing Framework:

1. **Initial Permutation (IP):** Incoming 64-bit plaintext blocks are systematically rearranged using a fixed structural permutation matrix.
2. **Block Division:** The permuted block splits into two symmetrical 32-bit segments: Left (L_0) and Right (R_0).
3. **16 Feistel Processing Rounds:** The system runs 16 successive algorithmic cycles. For each round (n), the transformation is defined as follows:
   • $L_n = R_{n-1}$ | • $R_n = L_{n-1} \oplus f(R_{n-1}, K_n)$ [where *f* represents the internal Feistel function and K_n is a specific 48-bit subkey].
4. **The Inner Feistel Function (*f*) Components:**
   • **Expansion (E-box):** Expands the 32-bit right-hand data vector to 48 bits.
   • **Subkey Mixing (XOR):** Combines the expanded 48-bit result with the round's unique 48-bit subkey.
   • **S-Box Substitution:** Routes the XOR output through 8 separate Substitution blocks (S-boxes), reducing the 48-bit string back to a dense 32-bit vector while introducing crucial non-linearity (confusion).
   • **Permutation (P-box):** Scrambles the 32-bit S-box output to provide diffusion.
5. **Final Swap & Inverse Initial Permutation (IP^{-1}):** After completing Round 16, the system swaps the final two halves, rejoins them into a single 64-bit structure, and processes it through the Inverse Permutation matrix to output the final ciphertext block.



22. Explain the integration of VPN with LDAP servers.

Architectural Integration Mechanics: Integrating a corporate VPN gateway with a **Lightweight Directory Access Protocol (LDAP)** server centralizes identity configuration management. Instead of creating and maintaining user accounts directly on individual local VPN components, authorization lookups are routed dynamically to a master enterprise index directory (like Microsoft Active Directory or OpenLDAP).

Step-by-Step AAA Interaction Sequence:

1. **User Connection Request:** A remote employee submits their unique enterprise corporate network credentials via their edge VPN client interface.
2. **Gateway Relay & LDAP Bind:** The VPN Gateway catches the request and opens a secure, encrypted connection (typically LDAPS over TCP port 636) to the destination LDAP directory node, executing an administrative lookup bind.
3. **Directory Attribute Match:** The LDAP directory processes the incoming validation query, locating the specific user object attribute string (e.g., `uid` or `sAMAccountName`) and assessing the password's validity.
4. **Profile Authorization Mapping:** Upon successful verification, the LDAP engine transmits the user's explicit profile metadata and security permissions (such as group strings: `memberOf: CN=Finance_VPN_Group`) back to the VPN Gateway.
5. **Access Enforcement:** The VPN Gateway parses the incoming organizational groups and applies the matching access rules, subnet restrictions, and routing rules to the user's active session.

23. Describe the types of VPN with examples.

1. **Site-to-Site VPN (Gateway-to-Gateway Architecture):** links permanent networks across different physical locations over public internet connections. Local network routers or perimeter firewalls act as dedicated VPN gateways handling all encapsulation and decryption. *Example:* A multinational medical enterprise connecting its primary data center in New York to regional clinics across Texas using persistent, static IPsec tunnels terminated on corporate Cisco ASA firewalls.
2. **Remote Access VPN (Client-to-Gateway Architecture):** links an individual isolated computer terminal safely to a secure internal enterprise backend network. Users initiate connection via a dedicated software app on their endpoint, which validates credentials and establishes an encrypted pipeline to the network edge firewall. *Example:* An accounting executive working from home launching a desktop application like FortiClient or Cisco AnyConnect to log in and securely access private internal payroll databases.
3. **SSL/TLS VPN (Browser-Centric Application Access Architecture):** provides fine-grained application-layer protection, typically requiring no dedicated local endpoint software client installation. Users gain protected access to specific internal company portals through standard web browsers running secure HTTPS sessions. *Example:* A human resources manager opening a standard Google Chrome web browser to securely pull records from a secure enterprise directory portal at `https://vpn.company.net/hr` without opening their entire computer up to the broader network infrastructure.